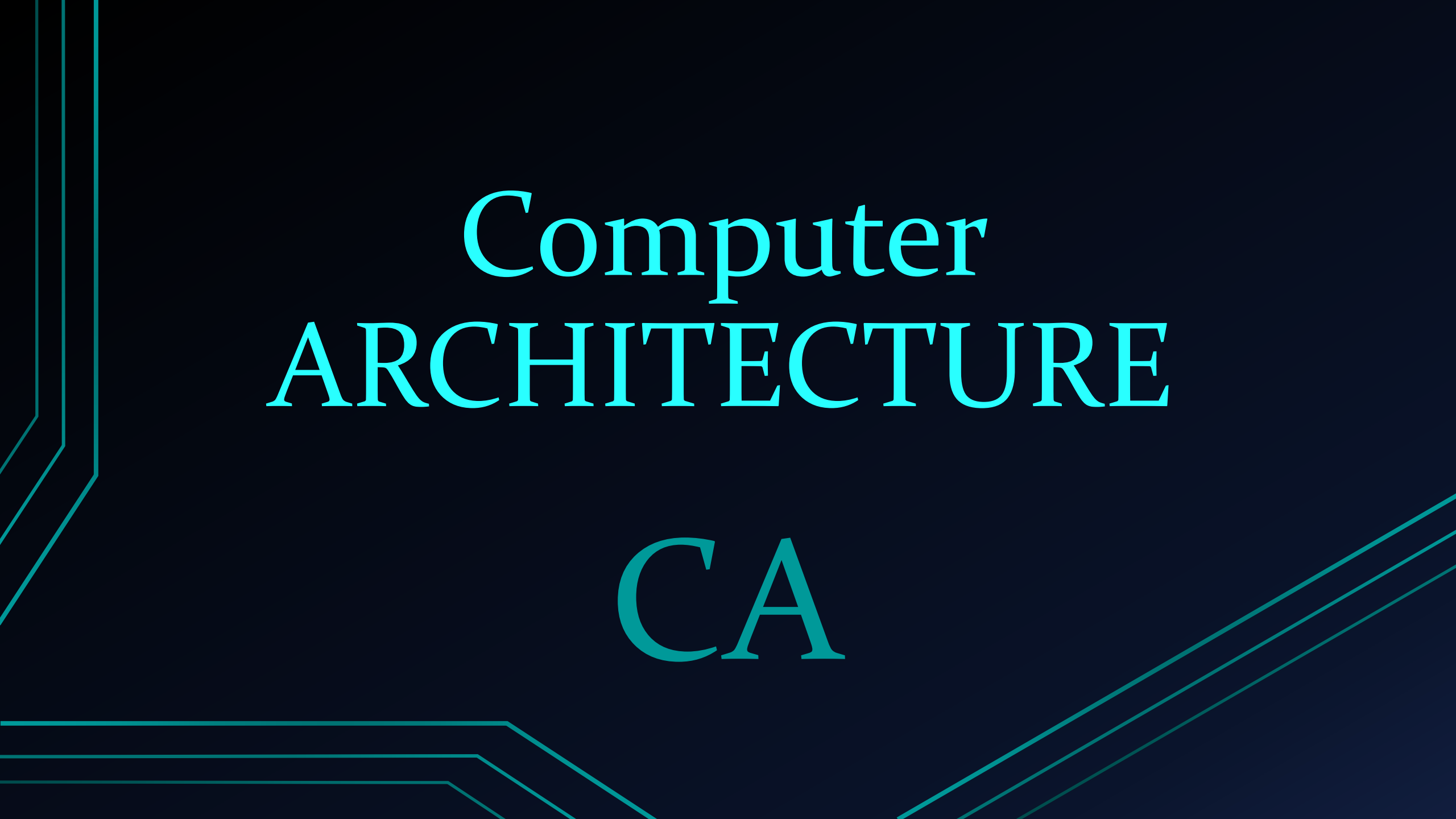




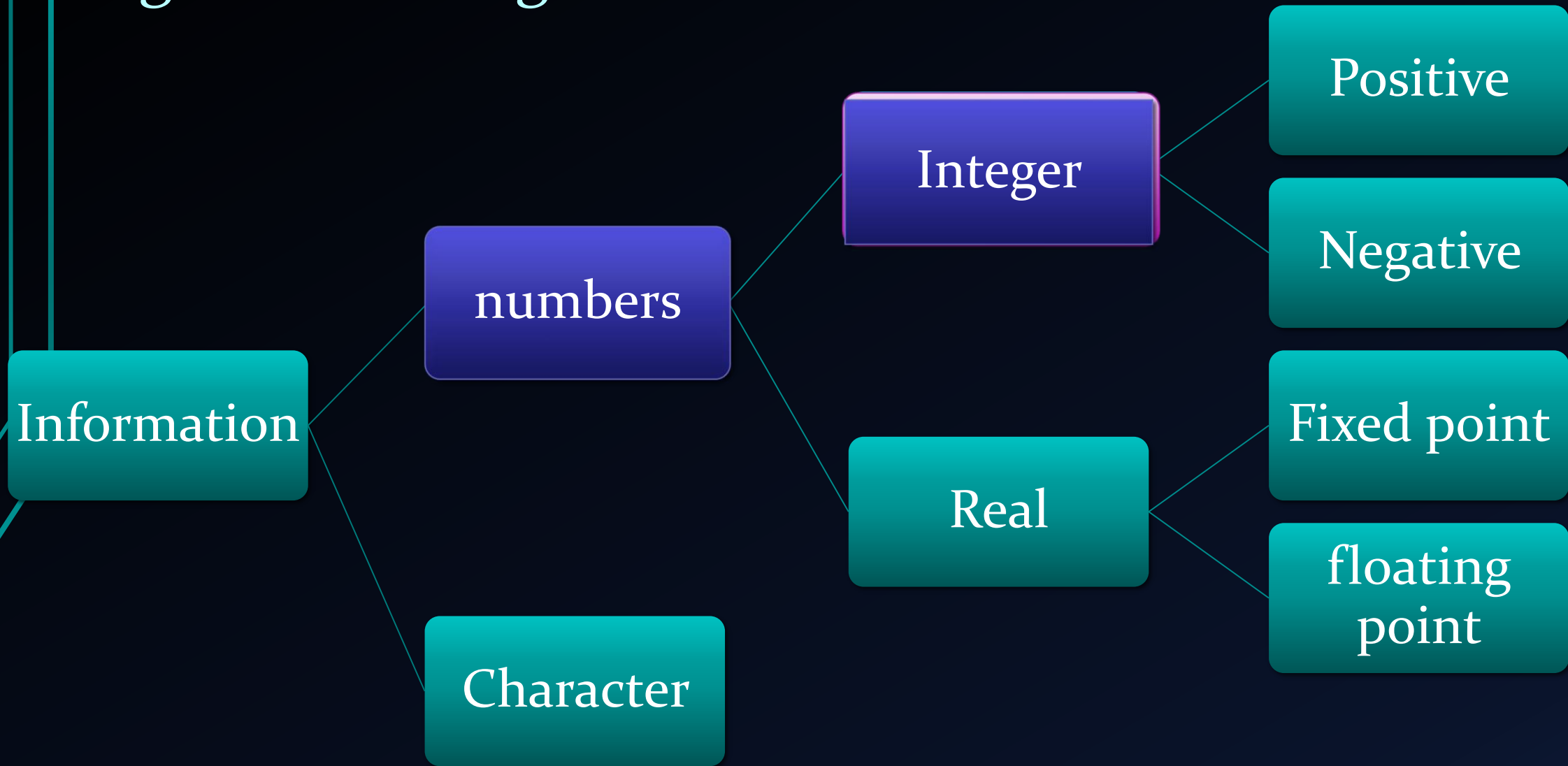
Computer ARCHITECTURE

CA

CHAPTER 1

NUMBER SYSTEMS

Digital encoding



Introduction

- We are accustomed to representing numbers using ten different symbols: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9.
This system is called the decimal system.
- Decimal numeration is the universal system of numeration today.
- This system represents numbers as a series of digits and sometimes includes a symbol, such as a decimal point or comma.
- **Example:** 2143; 21.2; ...

Number systems: Definition

- A number system is a writing system for expressing and representing numbers.
- It is a mathematical notation for representing numbers of a given set, using digits or other symbols in a consistent manner.

Number systems: Definition

- A number system is defined by:
 - **Its base (radix):** the number of distinct symbols used in the system.
 - **An alphabet A:** a set of symbols or digits used to represent values.
 - **Rules for writing numbers:** the arrangement of symbols in sequence to form numbers.
 - **Composition rules:** defining operations between values represented by symbols.
- The primary operations in a number system include addition, multiplication, and other arithmetic functions.

Number systems: Definition

- **Base B number system**
- An unsigned integer **N** can be represented in base B using n digits as follows:

$$N = (a_{n-1} + \dots + a_1 + a_0)$$

- In this format, known as positional representation, each digit a_i satisfies

$$0 \leq a_i \leq n$$

Number systems: Definition

- **Base B number system**
- Given a positive integer **N** , written in an **n** -digit base **B** , the polynomial form of **N** is as follows:

$$(N)_B = \sum_{i=0}^{n-1} a_i B^i$$

- $(N)_B = a_{n-1} \times B^{n-1} + \dots + a_1 \times B^1 + a_0 \times B^0$
- a_{n-1} is the **most significant digit** .
- a_i is the positional weight of the digit 'i'.
- a_0 is the **least significant digit** .

Number systems

- To avoid ambiguity, it is always essential to specify the base used.
- The notation $(Number)_b$ indicates that the number is written in base b
- Example:
 - The number $X = 100$ in base 2 is different from $Y = 100$ in base 10 and also from $Z = 100$ in base 16.
- $(100)_2 \neq (100)_{10} \neq (100)_{16}$



Some number systems

Number systems

- There are various number systems that use different sets of distinct symbols.
- In everyday life, we commonly use the decimal number system (base 10).
- In computing, we primarily use the binary number system (base 2).
- Other commonly used systems include the octal system (base 8) and the hexadecimal system (base 16)."

Decimal system

- This is the most commonly used numbering system today.
- The base is 10
- The alphabet is made up of ten numbers:
 $A = \{0,1,2,3,4,5,6,7,8,9\}$
- The decimal system is a positional numeral system
- The value of a digit depends on its position in the number.
- Numbers in a decimal system are written in the form
 - $(N)_{10} = a_{n-1} \times 10^{n-1} + \dots + a_1 \times 10^1 + a_0 \times 10^0$

Binary system

- The binary system is a positional number system
- The base is 2
- The alphabet is made up of two digits:

$$A = \{0,1\}$$

- The numbers in a binary system are written in the form
 - $(N)_2 = a_{n-1} \times 2^{n-1} + \dots + a_1 \times 2^1 + a_0 \times 2^0$
- Example : represent $(1101)_2, (100101)_2$
- $(1^3 1^2 0^1 1^0)_2 = 1 \times 2^0 + 0 \times 2^1 + 1 \times 2^2 + 1 \times 2^3$

Octal System

- The octal number system is a positional number system
- The base is 8
- The alphabet is made up of *eight* digits:

$$A = \{0,1,2,3,4,5,6,7\}$$

- Numbers in an octal system are written by
 - $(N)_8 = a_{n-1} \times 8^{n-1} + \dots + a_1 \times 8^1 + a_0 \times 8^0$
- Example : represent $(1298)_8, (128)_8$

Hexadecimal system

- The Hexadecimal number system is a positional number system
- The base is 16
- The alphabet is made up of sixteen numbers :
 $A = \{0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F\}$
- Numbers in an octal system are written by
 - $(N)_{16} = a_{n-1} \times 16^{n-1} + \dots + a_1 \times 16^1 + a_0 \times 16^0$
- Example : represent $(12AG)_{16}$

Decimal	Octal	Hexadecimal	Binary
0	0	0	0
1	1	1	1
2	2	2	10
3	3	3	11
4	4	4	100
5	5	5	101
6	6	6	110
7	7	7	111
8	10	8	1000
9	11	9	1001
10	12	A	1010
11	13	B	1011
12	14	C	1100
13	15	D	1101
14	16	E	1110
15	17	F	1111
16	20	10	10000

Base conversion

Base conversion

- Base conversion is the process of transforming the number N representation from a base B_1 to another base B_2 .
- In computer architecture, base conversion is essential for translating numbers between different storage formats. For instance, numbers are typically stored in binary format within memory. Base conversion is then used to translate these binary values into decimal format for user-friendly display.
- Write a number N from a base B_1 to another base B_2

$$(N)_{B_1} \equiv (N')_{B_2}$$

Conversion to decimal

- The method for converting numbers from binary, octal, hexadecimal, or another base to decimal relies on understanding that these systems are positional.
- In other words, each digit in binary, octal, hexadecimal, or another base has a positional weight, similar to the digits in the decimal system.

Conversion to decimal

- The decimal value of a number N written in a base B is obtained by its polynomial form;
- $(N)_B = (a_n a_{n-1} \dots a_0)_B$
- The decimal number is obtained as follows
- $(N)_{10} = a_n \times B^n + a_{n-1} \times B^{n-1} + \dots + a_0 \times B^0$
- The B^l is called 'quotient'

Converting decimal to base B

- The conversion algorithm:
- The rule to follow is successive division:
 - 1) Divide the decimal number by the target base b (using integer division);
 - 2) Take the result from step 1 and divide it by b again (integer division);
 - 3) Repeat this process until the result is zero;
 - 4) The sequence of remainders obtained represents the digits in the target base
- This process yields the least significant digit first, followed by the more significant digits as you proceed.

Converting decimal to base B

➤ Decimal to binary Conversion

- To convert a decimal number to binary, repeatedly divide the decimal number by 2 until the quotient becomes zero.
- This technique is known as the '**double-dabble**' method
- It involves recording the remainder at each division step.
- Finally, the remainders are read in reverse order, from bottom to top.

Converting decimal to base B

➤ Decimal to octal Conversion

- To convert a decimal number to octal, repeatedly divide the decimal number by 8 until the quotient is zero.
- This technique, known as the '**octal-dabble**' method,
- It involves recording the remainders at each step.
- Finally, the remainders are read from bottom to top, similar to the double-dabble method.

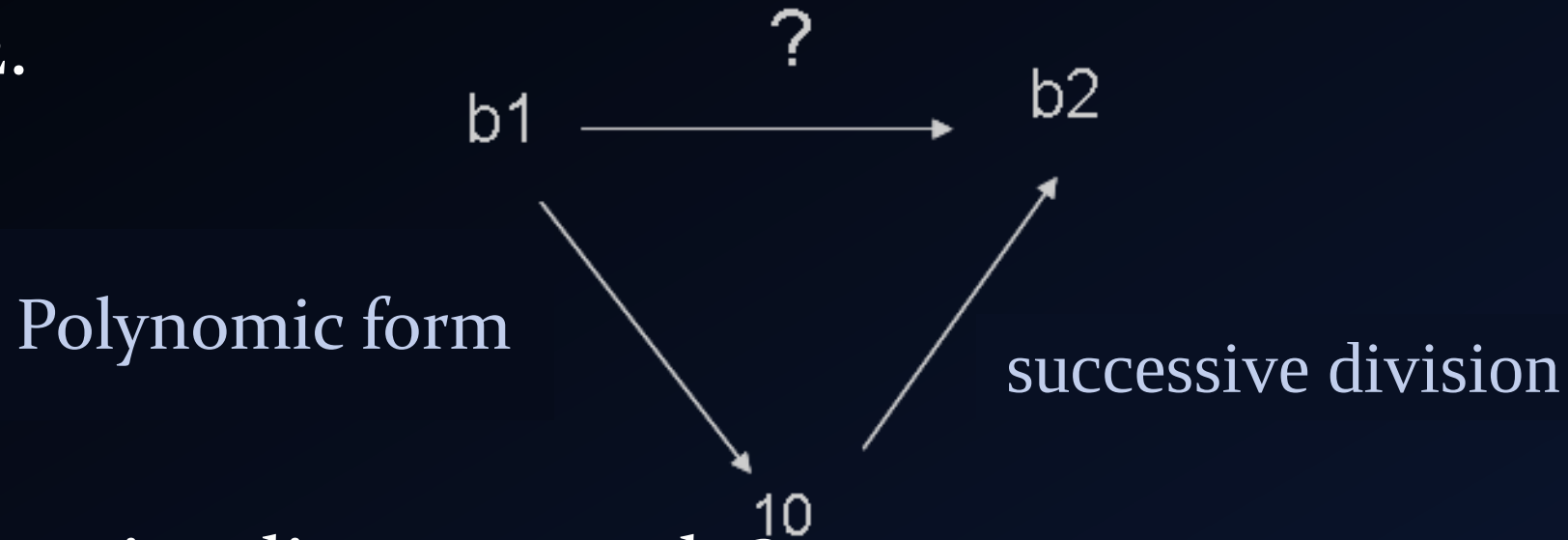
Converting decimal to base B

➤ Decimal to Hexadecimal Conversion

- The same procedure is followed to convert a decimal number to hexadecimal.
- Here, however, the decimal number is repeatedly divided by 16 until the quotient reaches zero.
- This method, known as '**hex-dabble**,' involves recording the remainder at each step.
- Finally, the remainders are read from bottom to top, just as in the previous methods.

Conversion from a B1 Radix to a B2 Radix

- There is no method to convert a number from base b_1 to another base b_2 directly.
- The general approach is to first convert the number from base b_1 to decimal, then convert the result from decimal to base b_2 .



- Is there a simpler approach ?

Conversion from a B₁ Radix to a B₂ Radix

- General solution:
 - Convert the number from B₁ base to decimal then from decimal to base B.
- Exceptions:
 - There is a simple method for base conversion between binary, octal and hexadecimal we replace each digit with the corresponding binary code.

Conversion from a B₁ Radix to a B₂ Radix

- From binary to Octal or hexadecimal:
- To convert a number N from binary to octal or hexadecimal we group the bits into subsets starting from the right, then replace them by its corresponding base symbol.
- From Octal or hexadecimal to binary:
 - To convert a number N from octal or hexadecimal to binary we

Conversion from binary to Octal

- Binary to octal :
- To convert from binary to octal we group the bits into subsets of three bits starting from the right.
- Each group is then replaced by its corresponding base-8 symbol.

Binary	000	001	010	100	101	110	111
Octal	0	1	2	3	4	5	7

Conversion from binary to Hexadecimal

- Binary to Hexadecimal:
- To convert from binary to hexadecimal we group the bits into subsets of four bits starting from the right.
- Each group is then replaced by its corresponding base-16 symbol.

Binary	0000	0001	0010	0100	0101	0110	0111
Hexadecimal	0	1	2	3	4	5	7
Binary	1000	1001	1010	1100	1101	1110	1111
Hexadecimal	8	9	A	B	C	D	F

Conversion from octal to binary

- Octal to Binary conversion:
- Each octal digit is represented as a 3-bit binary equivalent, and by combining all these binary groups, we obtain the final binary representation

000

Octal	0	1	2	3	4	5	7
Binary	000	001	010	100	101	110	111

Conversion from Hexadecimal to binary

- Hexadecimal to Binary conversion:
- Each octal digit is represented as a 4-bit binary equivalent, and by combining all these binary groups, we obtain the final binary representation

000

Hexadecimal	0	1	2	3	4	5	7
Binary	0000	0001	0010	0100	0101	0110	0111
Hexadecimal	8	9	A	B	C	D	F
Binary	1000	1001	1010	1100	1101	1110	1111

BINARY ARITHMETIC

Binary Arithmetic

- Binary addition
- The principles are the same as those used in base 10.
- Binary addition
- The addition of two binary bits is carried out according to the following specification:

X	Y	SUM	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Binary Arithmetic

- **The binary addition** of two numbers is carried out bit by bit from right to left (from the least significant bit to the most significant bit), carrying forward the carries.

Binary Arithmetic

- Binary subtraction
- The principles are the same as those used in decimal.
- **Binary subtraction** of two numbers is carried out like bit-by-bit addition from right to left (from the least significant bit to the most significant bit), carrying forward the carries.

Binary Arithmetic

➤ Binary subtraction

Binary subtraction			
X	Y	X - Y	Carry
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Binary Arithmetic

- Binary multiplication
- The principles are the same as those used in decimal.

Binary multiplication			
X	Y	X*Y	Carry
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	0

Binary Arithmetic

- **The binary multiplication** of two numbers is carried out from the least significant bit to the most significant bit, carrying forward the carries.

Binary Arithmetic

- Binary division
- The principles are the same as those used in decimal

The division			
X	Y	X/Y	Carry
0	1	0	0
1	1	1	0

- **The binary division** of two numbers is carried out from the most significant bit to the least significant bit